Autonomous Cognitive Engine for Deep Research and Long-Horizon Tasks

Overview

The Autonomous Cognitive Engine is a stateful, Large Language Model (LLM)–driven system designed to autonomously handle complex, long-horizon tasks such as deep research, multi-step reasoning, structured planning, and problem solving with minimal human intervention.

Unlike traditional chat-based systems, this engine plans, remembers, reasons, and executes actions over extended workflows using persistent memory and multi-agent collaboration.

Key Capabilities

- Structured task planning for long-horizon objectives
- Persistent memory using a Virtual File System (VFS)
- Supervisor–worker multi-agent delegation
- Stateful execution using LangGraph
- Model Context Protocol (MCP) based tool execution
- Observability and tracing with LangSmith
- Streamlit-based interactive user interface
- Automated testing with Pytest

System Workflow

1. User submits a complex request
2. Supervisor agent analyzes intent
3. Planning is triggered only when required
4. Sub-agents are delegated via MCP
5. Intermediate results are stored in VFS
6. Final response is synthesized and returned

Architecture Summary

User Input → Supervisor Agent → Planning Logic → Tool / Sub-Agent Execution
→ Memory Storage (VFS) → Final Output Generation

Project Folder Structure

src/

```
■■■ agents/
■ ■■■ chat_agent.py
■ ■■■ code_agent.py
■ ■■■ summarization_agent.py
■ ■■■ web_search_agent.py
■■■ memory/
■ ■■■ vfs_tools.py
■■■ tools/
■ ■■■ delegation_tool.py
■■■ utils/
■ ■■■ llm.py
■■■ tests/
■ ■■■ test_chatagent.py
■ ■■■ test_delegation_tool.py
■ ■■■ test_summarization_agent.py
■ ■■■ test_vfs_tools.py
■ ■■■ test_web_search_agent.py
■■■ main.py

LICENSE
README.md
requirements.txt
```

Technology Stack

- Python 3.10+
- LangGraph (stateful agent orchestration)
- LangChain (LLM and tool abstractions)
- Groq API (LLM inference backend)
- Model Context Protocol (MCP)
- LangSmith (tracing and observability)
- Streamlit (UI)
- Pytest (automated testing)
- python-dotenv (environment management)

Testing & Validation

The system includes automated unit tests for agents, delegation logic, and memory tools.
All test cases pass successfully, validating correctness and stability of the agent framework.

Project Status

Advanced Prototype

The system is stable, modular, and extensible. It supports autonomous planning, persistent memory, multi-agent collaboration, and is suitable for research-grade experimentation and future production hardening.